The Coordination Tax Is Paid in Fuel, Not Time: Light-Speed Delay in a Self-Replicating Probe Swarm

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Abstract*—A swarm of self-replicating probes fills a stellar field star by star, each probe settling a star and launching copies. The standard exploration-timescale model grants every probe perfect, instantaneous knowledge of which stars are already settled, and names finite light-speed as its own future work. We add that light-speed limit and measure its cost. The first finding is a caution: a model advanced in a timestep coarser than the interval between probe launches makes many probes decide from the same stale snapshot at once, and the resulting collisions appear to slow the fill by tens of percent. They do not. Resolved so that each probe decides at its true arrival time, light-speed lag does not measurably delay the fill – a large in-flight population keeps any single wasted trip off the critical path – and the gravitational-slingshot speed-up survives, reproducing the source model quantitatively at about 166 times powered flight. The real cost is redundant travel: with essentially the same number of probes built, stale views send probes to stars others have already claimed, a tax in fuel and effort, not time. It is governed by the probe's speed in units of light, from under a percent of journeys at slingshot speeds to about a fifth at the directed-energy speeds beamed propulsion targets; weighted by kinetic energy it is heavier still, several times the journey count for the fast slingshot policies and large even where the count tax is near zero. The tax grows with the replication branching factor and then saturates. It is not a hard floor: a probe that listens in flight, rather than only at its decision stars, recovers essentially all of the wasted-arrival tax at a small cost in fill time, so the finite speed of light sets how much coordination can ever buy back. The coordination cost a perfect-information model hides is paid in fuel, not time, and it scales with how fast the swarm flies.

***Index Terms*—self-replication, interstellar exploration, light-speed delay, multi-agent coordination, swarm dynamics**

I. Introduction

Aself-replicating probe does not explore a galaxy alone. It settles a star, builds copies of itself from local material, and sends them on to further stars, so a single launch grows into a front that sweeps outward and fills the reachable field. That a machine can build a copy of itself at all is von Neumann's result, and a probe of this kind saturating the Galaxy in a few million years is the sharpest form of the Fermi question: if interstellar settlement is possible it should already be complete, so the observed silence is a paradox [1], [2]. How long the filling takes is the exploration-timescale question, and it has a long modelling lineage. Early work framed each interstellar probe as an independent agent acting on priors fixed before launch [3], and treated settlement as a front: a density-dependent diffusion with a travelling-wave solution [4], or a Monte Carlo migration filling the Galaxy in tens of millions of years [5]. Later simulations made the field concrete, sending probes and subprobes through a defined stellar volume to measure fill times directly [6], [7], and established the seeded Monte Carlo methodology that situates an exploration timescale within a distribution of realizations rather than a single number [8].

The sharpest treatment for self-replicators is the slingshot line. Forgan, Papadogiannakis and Kitching showed that a single probe stealing energy from the motion of the stars it passes, a gravitational slingshot, explores up to two orders of magnitude faster than a probe under its own power [9]. Nicholson and Forgan extended that manoeuvre to self-replicating probes and confirmed the speed-up survives replication, with the surprising result that aiming for the nearest star can beat aiming for the largest boost [10]. The probe speeds that make this regime interesting, from the powered cruise of $3 \times 10^{-5}\,c$ up toward the 0.1 to $0.2\,c$ of directed-energy or high-velocity concepts [11], [12], are exactly where the next assumption starts to hurt.

That assumption, which Nicholson and Forgan name as a limitation and leave for future work, is that every probe is granted perfect, instantaneous knowledge of which stars are already settled. No probe ever flies to a star another has just claimed, because it always knows. In a real galaxy no such oracle exists: news that a star has been settled can travel no faster than light, so a probe choosing its next target acts on a picture of the field that is, for distant stars, years or centuries out of date. This is the elementary consequence of a finite signal speed, that events have only a partial ordering and a remote event is unknowable until its signal arrives [13].

This paper adds that light-speed limit to the self-replicating slingshot model and measures what it costs. The answer, built from a controlled paired-ensemble experiment, has two parts, and the first is a caution. The cost is easy to overstate: a simulation that advances in a fixed timestep coarser than the interval between probe launches makes many probes decide from the same stale snapshot at once, and they collide far more than they physically would. Read that way the delay appears to slow the fill of the field by tens of percent. It does not. When the dynamics are resolved so that each probe decides at its own arrival time, the slowdown vanishes: light-speed lag does not measurably delay the fill, and the slingshot speed-up survives contact with finite information intact, reproducing the source model quantitatively. The real cost shows up elsewhere. Every star still gets settled, and with a negligible change in how many probes are built, well under a percent; what the delay changes is how many wasted journeys the swarm flies to stars other probes have already claimed. That is a cost in fuel and effort, not in time, and a perfect-information model hides it entirely. It is governed cleanly by a single dimensionless number, the probe's speed in units of the speed of light: negligible for a powered cruise, about a percent of journeys at slingshot speeds, and rising to roughly a fifth at the directed-energy speeds that beamed-propulsion concepts target. We show it is heavier still when weighted by the energy each wasted hop cost, that it grows with the replication branching factor before saturating, and – because a probe deciding only at its stops is pessimistic – that a probe which listens in flight recovers almost all of it, which bounds how much coordination can ever buy back. The coordination tax is paid in fuel, not time, and it scales with how fast the swarm flies.

II. The light-delayed belief model

A. The knowledge gate

The change to the model is one gate on knowledge. A settled star is treated as an omnidirectional beacon that begins emitting in the year it is settled. A probe deciding its next target at star f in year Y believes a distant star i is settled only once that beacon's light has had time to arrive:

$$\mathrm{settled}(i) + \frac{\mathrm{dist}(f,i)}{c} \leq Y. \tag{1}$$

Under the perfect-information assumption of the source model, (1) collapses to "settled at any non-negative year," and the model reproduces our perfect-information baseline bit-for-bit; the light-speed term is what introduces stale views. When two probes both believe an unsettled star is the best target they both fly to it; the loser arrives to find it taken, its trip wasted, and must re-target from the even more stale picture it holds on arrival. This is availability chosen over consistency: rather than wait for a single agreed map that finite light-speed makes impossible, each probe acts now on its local view [13]. Stars carry velocity magnitudes that feed the slingshot boost but are held fixed in position, following the source model, so $\mathrm{dist}(f,i)$ in (1) is static and the beacon geometry does not shift under the probe. Physical arrival always reads ground truth: what a probe finds on arrival is the star's real status, whatever it believed when it aimed.

B. The three flight policies, and the speed axis

We carry three named policies, following the slingshot studies we extend [9], [10]. *Powered nearest-neighbour* flight moves under the probe's own power to the nearest unsettled star. *Nearest-star slingshot* flight takes the same nearest target but gains speed from gravitational slingshots off the stars it passes. *Maximum-boost slingshot* flight instead selects, among the nearest believed-unsettled stars, the target that yields the largest velocity gain. The gravitational boost self-limits (its per-encounter gain falls off once the probe outruns the stellar escape speed), so both slingshot policies saturate at a mean speed of a few thousand km/s, a few times 10^{-3} of c. To reach the higher speeds that beamed directed-energy concepts target, 0.1 to $0.2\,c$ [11], [12], a probe must be driven, not slung; we therefore sweep the powered cruise speed as a clean, direct control on v/c, from the source model's $3 \times 10^{-5}\,c$ up to $0.2\,c$. Powered flight thus spans the whole speed axis, while the two slingshot policies sit near $v/c \approx 0.01$.

C. Event-driven resolution, and why it is necessary

The fold can advance either by a fixed timestep or by jumping to the next probe arrival. This is not a mere numerical convenience, and getting it wrong is the single largest source of error in this problem. A fixed timestep Δt processes together every probe that arrives within a window, so all of them decide against the same snapshot of the settled map. When Δt exceeds the interval between launches – which it does badly in the boosted regime, where a probe crosses a parsec in centuries against a source-model step of thousands of years – the window batches many independent decisions into one synchronized burst, and they collide far more often than they would if each decided at its true arrival time. The measured coordination cost then reflects the size of the timestep, not the physics. We therefore run the model event-driven throughout: the fold jumps to the earliest outstanding arrival, advances the clock to it, and processes exactly the probes arriving then. This is the exact $\Delta t \to 0$ limit, dt-independent by construction, and it is what every number in this paper is computed on. Section III shows directly that the apparent fill-time tax is an artifact of a coarse step and collapses to zero as the step resolves.

D. The governing ratio

How much the gate matters is governed by one dimensionless number: the information delay across a hop divided by the travel time across it. For a hop of length d the delay is d/c and the travel time is d/v, so the ratio is

$$\Lambda \;=\; \frac{d/c}{d/v} \;=\; \frac{v}{c}, \tag{2}$$

the probe's speed in units of c, independent of the hop length. A slow probe ($\Lambda \ll 1$) outpaces its own ignorance only trivially, so stale views barely arise; a fast probe carries a large Λ and routinely decides before the relevant news can reach it. At the powered cruise speed $\Lambda \approx 3 \times 10^{-5}$ is negligible. As we show below, once the dynamics are resolved Λ does not predict a fill-time penalty (there is essentially none), but it *does* govern the redundant-travel penalty cleanly and monotonically – it is the right parameter, for the right cost.

E. Experimental design and parameters

The field is a box of stars at a uniform density of one star per cubic parsec, the convention of the source model. Each settled star launches offspring (the replication branching factor): branching is what makes two probes ever contend for a star, so it is the precondition for any coordination effect at all and the parameter most likely to change the answer, so we take two offspring as the default and sweep it explicitly (Section III-E). Three properties keep the measurement honest. First, knowledge is evaluated only at the decision star at the moment of decision; news a probe's path happens to cross in mid-flight is ignored, which deliberately undercounts what a probe could know and so makes the measured waste a conservative upper bound – we measure the opposite, optimistic bound (a probe that listens in flight) separately in Section III-F. Second, a probe that loses a race re-targets, but only up to a fixed number of times before it is retired; this cap is applied identically under both coordination regimes, so the paired comparison is fair, and the measured tax converges as the cap is raised (rising while the cap still truncates live bounce chains, then saturating), so the near-converged default is a mild lower bound rather than a free knob. Third, the whole simulation is a pure, seeded, event-driven fold: fixing the seed fixes the star field, every arrival, and every target choice, so a run with the light-speed gate and a run without it can be compared on the *identical* galaxy. That paired design attributes a difference to the delay alone rather than to the luck of the star layout.

Two dependencies are worth stating because they are easy to over-read. Changing the value of the uniform density rescales every inter-star distance by a common factor, multiplying travel time and light-travel time on every hop equally, so Λ and the relative penalties are unchanged (a pure geometric rescaling of the absolute clock); we confirm this holds to the printed digit across densities from the sparse solar neighbourhood [14] near 0.14 up to 5 stars per cubic parsec. And because the field always fills under both regimes and each settlement launches a fixed number of offspring, the two regimes build essentially the same number of probes – they differ only in a handful of terminal launches as targets run out, well under a percent – so the resource the delay actually moves is the number of journeys flown, which is why the coordination cost is a travel cost.

F. Baseline validation

By construction the "instant" regime is the $c \to \infty$ limit of (1), and it reproduces the plain perfect-information fold bit-for-bit. Against Nicholson and Forgan's published model [10] the event-driven fold reproduces both headline findings *quantitatively*: gravitational slingshots explore about two orders of magnitude faster than powered flight (a factor of about 166 on a 400-star field), and their surprising result that aiming for the nearest star beats aiming for the largest boost on total exploration time holds, while the maximum-boost policy reaches the higher peak speed. A coarse fixed step had quantized the boosted hops and undercut this speed-up to about twenty-fold; resolving the step recovers the full factor, which is the same correction that removes the spurious fill-time tax.

III. Results: the coordination tax

We measured the cost as a paired ensemble: 48 seeded galaxies of 500 stars, each filled twice on the *identical* field, once with perfect information and once with the light-speed gate of (1), all at the resolved event-driven timestep. Reporting the distribution across the ensemble follows the seeded Monte Carlo practice of the field [8]; we give the median with a seeded percentile-bootstrap 95% confidence interval. We deliberately do not lean on a significance test: by the structure of the model a stale view can only make a probe believe a settled star is still open, never the reverse, so the light-delayed run can essentially only *add* wasted journeys. The sign of the effect is therefore built in; what is informative is its magnitude and how it scales, which is what the intervals and the sweeps below report.

A. There is no fill-time tax; the coarse-timestep one is an artifact

Light-speed lag does not measurably slow the fill of the field. Fig. 1 shows why the literature intuition says otherwise. Run with the source model's fixed timestep the fill-100% penalty for nearest-star slingshot flight is a median of 30.3%, positive in all 32 seeds – a large, robust-looking tax. It is an artifact of the step. As the timestep is refined the penalty falls monotonically – 27.6%, 25.0%, 12.8%, 5.3% at 2000, 1000, 500, 250 years – and at the event-driven limit it is a median of 0.0%, no longer distinguishable from zero. The coarse step had batched many launches into one window so they decided from the same stale snapshot and collided; resolving the step lets each probe decide at its own arrival time, and the collisions that cost time disappear. The same refinement restores the slingshot speed-up to its full value: against powered flight the event-driven fold reproduces Nicholson and Forgan's roughly two-orders-of-magnitude figure (about $166\times$ on a 400-star field), where the coarse step had reported only about $20\times$. The celebrated speed-up survives contact with finite information; the fill-time tax does not exist.

B. Why wasted travel costs no time

The reason a wasted journey costs fuel but not time is that the front is carried by a large parallel population, and a loser is only ever one member of it. Fig. 2 tracks the number of probes in flight as the field fills: exponential branching keeps a median of about 480 probes aloft at the peak of a 500-star fill, and still about 440 in flight when the field is 90% settled. Crucially the light-delayed swarm carries essentially the same in-flight population as the perfect-information one (peaks of about 470 against 480), because both build the same probes and fill on the same schedule. A probe that loses a race and flies on to a claimed star is thus one of several hundred in transit; the last star is reached by whichever probe is nearest and soonest, and a loser's redundant trip does not sit on that critical path. Removing the wasted trip, or adding it, moves fuel but not the finishing time. This is the concrete mechanism behind the null result of the previous subsection.

TABLE I

COORDINATION TAX OVER THE 48-SEED PAIRED ENSEMBLE AT THE EVENT-DRIVEN TIMESTEP, POWERED FLIGHT, AS A FUNCTION OF $\Lambda = v/c$: THE REDUNDANT-TRAVEL (FUEL) TAX AS A PERCENT OF THE WASTED JOURNEYS UNDER PERFECT INFORMATION, AND THE FILL-TIME TAX, EACH AS A MEDIAN WITH A BOOTSTRAP 95% CONFIDENCE INTERVAL.

$\Lambda = v/c$	Fuel tax (%)	Fill-time tax (%)
0.01	0.6 [0.5, 0.8]	0.0 [0.0, 0.0]
0.03	2.9 [1.8, 4.1]	0.1 [0.0, 1.3]
0.05	4.2 [3.4, 5.8]	0.3 [0.0, 1.8]
0.10	9.3 [7.4, 11.4]	2.6 [0.9, 4.0]
0.20	19.5 [17.8, 23.3]	6.6 [5.2, 10.5]

C. The real cost is redundant travel, governed by $\Lambda = v/c$

What the delay changes is how far the swarm flies. Because the field always fills and each settlement launches a fixed number of offspring, both coordination regimes build essentially the same number of probes – differing by well under a percent, in a handful of terminal launches – so the resource the delay moves is journeys, and stale views add wasted ones. That redundant-travel tax is governed cleanly by the probe's speed in units of c. Table I and Fig. 3 give it for powered flight swept across the speed axis: negligible at the cruise speed, a median of 0.6% of journeys at $\Lambda = 0.01$, rising monotonically to 19.5% at $\Lambda = 0.2$. The effect is not marginal – its confidence interval clears zero at every speed, and Fig. 4 shows the per-seed distributions sitting above zero and widening with Λ. A small fill-time tax rides alongside it once the swarm is fast, growing from nil at $\Lambda \leq 0.01$ to a few percent at $\Lambda = 0.2$ (Table I), but the redundant travel is the larger and cleaner signal, and unlike the coarse-step time tax it is real: it does not vanish when the dynamics are resolved, because it is not a batching artifact but the physical consequence of deciding on stale views.

This is what Λ was always meant to govern. The ratio is the same on every hop, so it cannot pick out where in the field the cost lands; but it sets, cleanly and monotonically, how much of the swarm's motion is wasted. It is the right dimensionless group for the fuel tax, and it does not predict a time tax because at resolved dynamics there is essentially none to predict.

D. The tax is heavier weighted by energy

Counting journeys understates the cost, because a fast wasted journey squanders far more than a slow one: the kinetic energy to reach speed v scales as v^2, so weighting each wasted trip by the specific energy $\frac{1}{2}(v/c)^2$ its launch cost gives the true burden. (The relativistic form $(\gamma - 1)c^2$ exceeds this Newtonian weight by under 0.8% up to $0.1\,c$ and only 3.1% at $0.2\,c$, so the Newtonian figure is a safe primary metric.) The weighting matters most where journeys travel at different speeds, which is exactly the slingshot regime. Fig. 5 compares the count tax with the energy-weighted tax for each policy. For powered flight, where every journey is at one speed, the two coincide. For nearest-star slingshot flight the count tax is only 0.8%, but because its wasted trips fly at about 2,700 km/s the energy tax is 4.2%, and up to 8.4% once one accounts for the deceleration a rendezvous requires. For maximum-boost flight the gap is starkest: the count tax is essentially zero, yet the energy tax is 11.3 to 22.6%, because that policy's few wasted trips are its fastest and farthest. We report the energy tax as a band between a flyby, which pays the one-way acceleration energy, and a rendezvous, which must brake to rest and re-accelerate and so pays about twice; beamed light sails in particular arrive with no braking beam and must carry their own deceleration [11]. The lesson is that a policy can hide its coordination cost in the journey count and still pay it in energy.

E. Contention scales with the branching factor

The tax exists only because probes contend for stars, and contention is created by replication: each settlement launches offspring, and it is those offspring that can race. The branching factor is therefore the parameter most likely to change the answer, so we sweep it. Fig. 6 shows the fuel tax for two, three and four offspring per settlement. At moderate speed ($\Lambda = 0.05$) it is flat, near 6% regardless of branching. At directed-energy speed ($\Lambda = 0.2$) it grows with branching, from 18.4% at two offspring to 24.9% at three, then saturates at 25.5% at four: more offspring make more simultaneous races, but once enough probes are already contending the marginal collision rate levels off. The default of two offspring is thus not a floor that hides a much larger tax at higher branching; the tax rises modestly and then plateaus, and the whole effect stays within the same order of magnitude.

F. The physical floor: how much survives relaying in flight

The tax reported so far is a conservative upper bound, because a probe here learns only at its decision stars and ignores news its path crosses in flight. The opposite extreme is a probe that listens continuously: the moment the beacon from a now-claimed target overtakes it – at the closed-form time set by the beacon closing on the probe at c while the probe closes on the target at v – it abandons the doomed hop and re-aims from wherever it then is. This "in-flight relay" mode is the optimistic bound on what any relay protocol can recover, and we measure it directly. Its effect is dramatic. Because a beacon travelling at c always overtakes a probe travelling at $v < c$ before the probe arrives, a listening probe essentially never completes a wasted arrival: the count of wasted arrivals falls from about 2,070 under decision-site knowledge to zero at $\Lambda = 0.2$. Redundant travel falls too, and below even the perfect-information baseline – from about 3,580 pc under the light-speed gate, and 2,930 pc under perfect information, to about 1,640 pc – because listening in flight also lets a probe bail out of an "assignment collision" (two probes independently choosing the same open star) that perfect knowledge at the decision site does not prevent. The recovery is not free: the mid-flight detours lengthen the fill slightly, by about 2% at $\Lambda = 0.2$. So the decision-site tax is an upper bound, not a floor: a swarm that relays in flight can

TABLE II

WHERE REAL PROPULSION CONCEPTS SIT ON THE Λ AXIS, AND THE FUEL TAX EACH INCURS. THE POWERED CRUISE IS FAR TOO SLOW FOR LAG TO BITE; GRAVITATIONAL SLINGSHOTS SELF-LIMIT NEAR $\Lambda \approx 0.01$; ONLY BEAMED DIRECTED-ENERGY PROPULSION REACHES THE REGIME WHERE THE TAX IS LARGE.

Propulsion regime	$\Lambda = v/c$	Fuel tax (%)
Powered cruise [10]	3×10^{-5}	≈ 0
Gravitational slingshot [10]	≈ 0.01	≈ 1
Directed-energy / light sail [11]	0.1 to 0.2	9 to 20

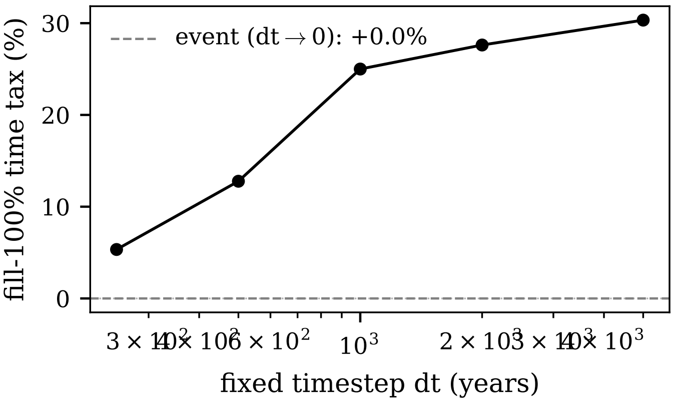

Fig. 1. The fill-100% time penalty for nearest-star slingshot flight versus the fixed timestep, over the 32-seed ensemble. The apparent penalty is monotone in the step and collapses to zero (dashed line: the event-driven, dt-independent limit). The tens-of-percent "coordination tax" of a coarse-step model is a discretization artifact, not a physical slowdown.

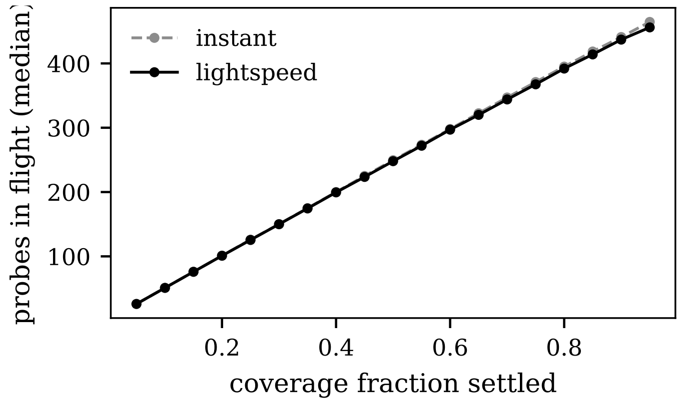

Fig. 2. Probes in flight versus the fraction of the field settled, for perfect information and the light-speed gate (medians over 16 seeds, 500-star field, $\Lambda = 0.2$). Hundreds of probes are aloft throughout the fill and the two regimes track each other closely, so a single wasted journey is never on the critical path to the last star – the mechanism behind the absence of a fill-time tax.

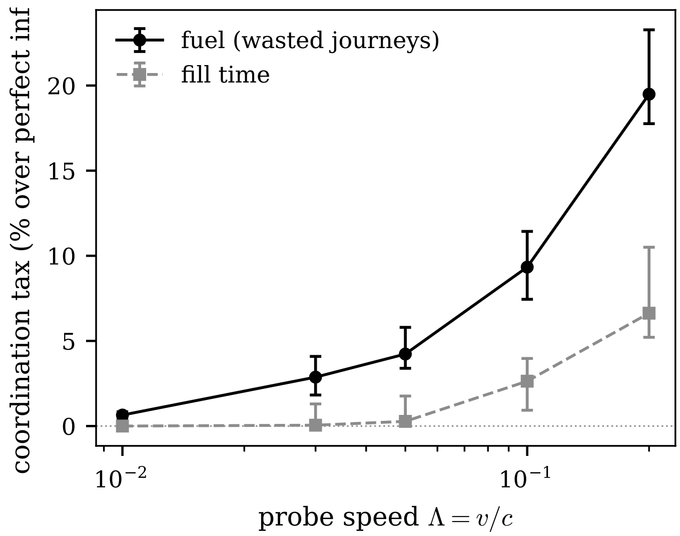

Fig. 3. The coordination tax versus probe speed $\Lambda = v/c$, powered flight, over the 48-seed ensemble at the event-driven timestep (points: medians; bars: bootstrap 95% confidence intervals). The redundant-travel (fuel) tax rises cleanly and monotonically with Λ, from under a percent at slingshot speeds to about a fifth at directed-energy speeds; the fill-time tax is a small companion that only appears once the swarm is fast.

recover essentially all of the wasted-arrival tax and some of the collisions perfect information leaves behind, at a small cost in time. What remains as a true floor is the partial travel a probe flies toward a doomed star before the beacon reaches it, which grows with Λ as the probe covers more of the hop before the relatively slower news catches up.

G. *Where the cost falls*

Table II places the physical propulsion concepts on the speed axis. The source model's powered cruise ($3 \times 10^{-5}\,c$) is far too slow for any race to matter, and pays nothing. The gravitational slingshot that motivated this study reaches only $\Lambda \approx 0.01$ before its boost self-limits, and pays about a percent of journeys – real, but small (and, as Section III-D shows, several times that in energy). The tax becomes large only for beamed directed-energy or light-sail propulsion at 0.1 to $0.2\,c$ [11], [12], where it takes a tenth to a fifth of journeys. Every case still fills the field to completion: this is a property of the model, not a result, since with re-targeting and no loss term the reachable field must close, and delay supplies no term that would leave a permanent unsettled remainder of the kind an "Aurora" steady state requires [12]. The coordination failure shows up as wasted motion, not as an unfilled galaxy.

IV. DISCUSSION

The result refines rather than overturns the slingshot picture, and it does so twice over. Nicholson and Forgan's speed-up is real, large, and here reproduced quantitatively [10]; finite light-speed does not erode it, and does not measurably delay the fill of the field. What a perfect-information model hides is not lost time but lost motion: probes flying to stars that others have already claimed. That redundant travel is the coordination tax, it is governed by the probe's speed in units of c, and it matters only once a swarm flies fast enough for a decision to outrun its own news.

A. *Why the cost is fuel*

Because the field always fills and each settlement launches a fixed number of offspring, the two coordination regimes build essentially the same number of probes, differing by well under a percent; the delay changes almost nothing about how much

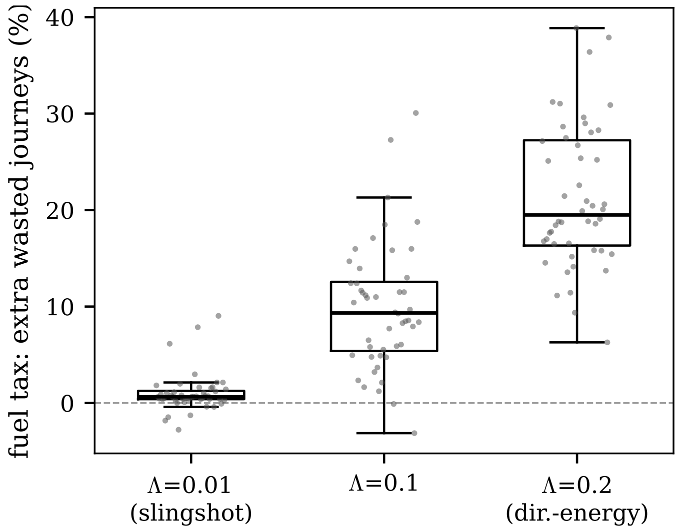

Fig. 4. Per-seed distribution of the fuel tax (extra wasted journeys, as a percent of the perfect-information waste) at slingshot speed ($\Lambda = 0.01$), $\Lambda = 0.1$, and directed-energy speed ($\Lambda = 0.2$), over the 48-seed ensemble (box: interquartile range and median; points: individual seeds). The tax is small but consistently positive at slingshot speed and unambiguous at directed-energy speed.

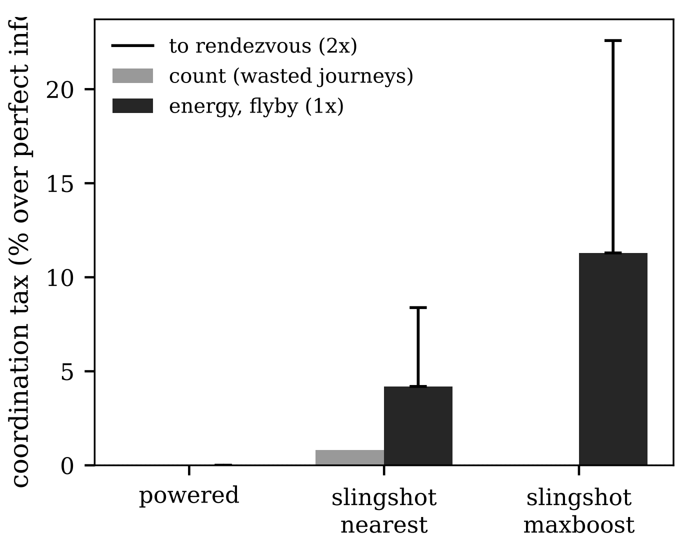

Fig. 5. Coordination tax by flight policy, counted as wasted journeys (grey) and weighted by the kinetic energy each wasted journey cost (black bar: flyby, one-way acceleration; whisker: up to a rendezvous that must brake and re-accelerate), over the 32-seed ensemble. For the slingshot policies the energy tax far exceeds the count tax, because their wasted trips are their fastest.

is manufactured, only how far it is flown. The tax is therefore a travel cost. In energy terms it is heavier than the journey count suggests, and increasingly so with speed, since the kinetic cost of a hop scales with v^2; Section III-D makes this quantitative and finds that for the slingshot policies the energy-weighted tax is several times the count tax, and can be large even where the count tax is near zero. This is the sense in which the tax "scales with how fast the swarm flies": both the rate of wasted trips and the cost of each rise with v.

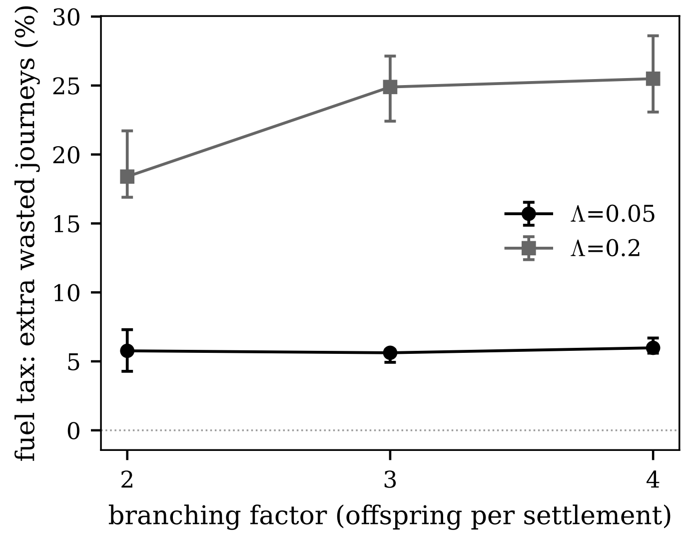

Fig. 6. Fuel tax versus the replication branching factor (offspring per settlement) at moderate and directed-energy speed, over the 32-seed ensemble (points: medians; bars: bootstrap 95% confidence intervals). Contention grows with branching at high Λ and then saturates; the default of two offspring does not hide a much larger tax.

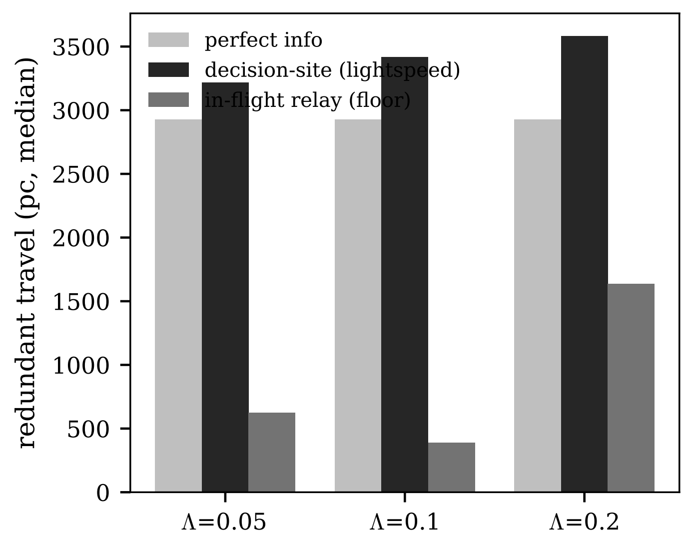

Fig. 7. Redundant travel under perfect information, the decision-site light-speed gate, and in-flight relay, versus Λ (medians over 48 seeds, 400-star field). Listening in flight drives completed wasted arrivals to zero and cuts redundant travel below even the perfect-information baseline, recovering the light-lag tax at a small cost in fill time.

B. Limitations

The decision-site headline is a conservative upper bound: probes here learn only at their decision stars, ignoring news their paths cross in flight. We relax exactly that assumption in Section III-F, where a probe that listens in flight recovers essentially all of the wasted-arrival tax, so the headline is an upper bound rather than a floor. A true probe-to-probe relay of the settled map, which we do not implement, would sit between the beacon-only in-flight bound and the decision-site one and is the natural next step. The floor beneath all of them

– that no probe can know a distant star is taken until its light arrives – is set by physics, not protocol.

A further, inherited idealization concerns the stars themselves. They carry the velocity vectors that power each slingshot but are held fixed in position, exactly as in the two source models, both of which name this as their single most important simplification [9], [10]. It is not a small one: at the local circular speed a star sweeps about 225 pc in a million years, and even the peculiar velocity dispersion alone carries it some 30 to 40 pc [15], [16], far past the roughly one-parsec mean hop, so over a Myr-scale fill real proper motion would reshuffle the beacon geometry the knowledge gate reads. The effect is bounded for our result rather than fatal to it: both coordination regimes run on the *identical* frozen field, so the paired difference still isolates the light-lag cost even where the absolute geometry is idealized – but the absolute fill times inherit the idealization.

Three further bounds on scope. First, the field is uniform; changing its density only rescales the clock (confirmed), but a clumpier distribution would lengthen the long-range hops and is a genuine unknown we do not probe. Second, at high v/c the hops are short, so an event-driven run generates of order one arrival per star settled many times over and costs roughly $O(N^2)$; this bounded the size sweep to a sixteen-fold range of field sizes, over which the fuel tax holds as a stable fraction. We therefore report scale-stability across the tested range only, and make no quantitative extrapolation to galactic ($\sim 10^{11}$-star) scale – a sixteen-fold lever arm cannot support one. Third, the dynamics have no loss term: settled sites never fail, so the field always closes. A settlement-death term of the kind that produces the "Aurora" steady state [12] is what would be needed to turn coordination failure into a permanent unsettled remainder; delay alone does not.

C. The signature of delayed agreement

The behaviour is, by analogy rather than derivation, the expected signature of delayed multi-agent agreement. Formal consensus degrades as the ratio of communication delay to decision cadence grows: the standard protocol is stable only while the one-hop delay stays below a bound set by the coupling [17], generalized later to switching topologies and link failures [18]. We do not apply that bound quantitatively – it needs a probe decision cadence the literature does not fix, and it does not predict the $\Lambda = v/c$ scaling that is this paper's actual result – so we offer it as context rather than derivation; but the regime is unambiguous: at a parsec spacing the round-trip light-time is years, far longer than any plausible cadence, so the probes sit deep in the region where guaranteed agreement is not merely slow but impossible [19], and a system that must keep acting trades a single consistent map for availability [20], which is exactly what (1) encodes. The same progression drove the space teleoperation and networking communities from closed-loop control through move-and-wait and supervisory control [21], [22] to hop-by-hop delay-tolerant store-and-forward once no continuous path survives [23]–[25]. At interstellar hops every faster regime is left behind; each probe is an independent colony acting on priors, and the coordination tax is what that independence costs in fuel when probes want the same stars.

D. A design lesson

The corollary is a threshold, not a preference. At slingshot speeds the tax is about a percent of journeys: coordination machinery – relay, gossip, a shared map – is not worth its mass, and a designer should spend everything on speed. At the directed-energy speeds that make a probe fast enough to matter, the same physics that buys the speed also makes stale views expensive, and a fifth of journeys wasted is worth engineering against. The faster the swarm you can build, the more a little coordination is worth buying – and the finite speed of light sets how much you can ever buy back.

V. Conclusion

Adding a finite speed of information to the self-replicating slingshot model turns a convenient assumption into a measurable cost, but not the cost a coarse model reports. Once the dynamics are resolved so that each probe decides at its true arrival time, light-speed lag does not measurably slow the fill of the field, and the slingshot speed-up survives intact, reproducing Nicholson and Forgan quantitatively; the tens-of-percent fill-time penalty seen in a fixed-timestep model is a discretization artifact. The genuine cost is redundant travel. Every star is still settled, with a negligible change in the number of probes built, but stale views send probes to stars other probes have already claimed, and that waste is governed cleanly by the probe's speed in units of the speed of light, $\Lambda = v/c$: negligible for a powered cruise, about a percent of journeys at slingshot speeds, and roughly a fifth at the directed-energy speeds beamed propulsion targets. Weighted by the kinetic energy each wasted hop cost it is heavier still, and for the fast slingshot policies it is large even where the raw journey count barely moves; it also grows with the replication branching factor before saturating. The coordination tax a perfect-information model hides is paid in fuel, not time, and it scales with how fast the swarm flies. It is not a hard floor: a probe that listens in flight, one of the two extensions the model invites, recovers essentially all of the wasted-arrival tax at a small cost in fill time, while a full probe-to-probe relay of the settled map remains open. Neither can remove the physical floor beneath the tax, and how much either can buy back is itself bounded by the finite speed of light.

Data Availability

The model is a deterministic, seeded, event-driven fold; every figure, table, and reported statistic regenerates from source by running the swarm module's experiment and figure scripts, and every quantitative claim traces to a cited source in the project's shared bibliography. The code and its citation metadata are archived with the concept DOI `10.5281/zenodo.21249296`.

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